Natural Language Processing

Class 7: Classification Tasks

Adam Faulkner

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1. Introduction
2. Information Extraction
3. Relation Extraction
4. Coreference Resolution
5. Logical Entailment
6. Intent Classification
7. Argumentation Mining

1 Introduction

2 Information Extraction

3 Relation Extraction

4 Coreference Resolution

5 Logical Entailment

6 Intent Classification

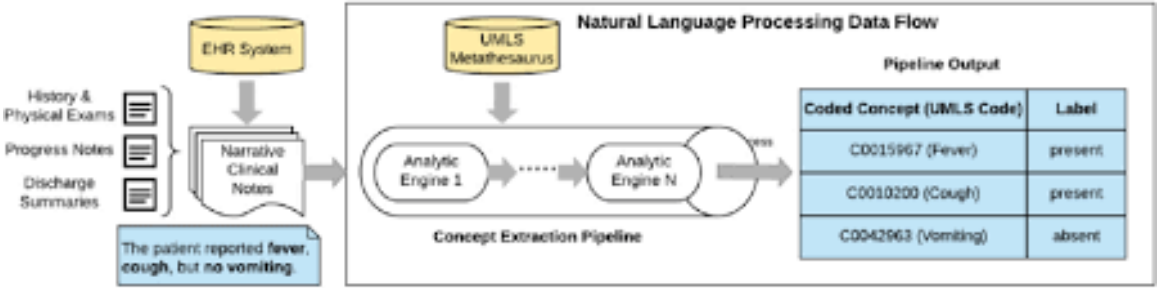
7 Argumentation Mining

Classifiers are the backbone of the classic NLP pipeline

- We covered generic classification in Class 1
- But classification is used everywhere in the classic NLP pipeline
- Let's take a look at a standard, pipeline-based approach to identify the presence of symptoms in clinical notes
- There are multiple steps in this pipeline that require classifiers...

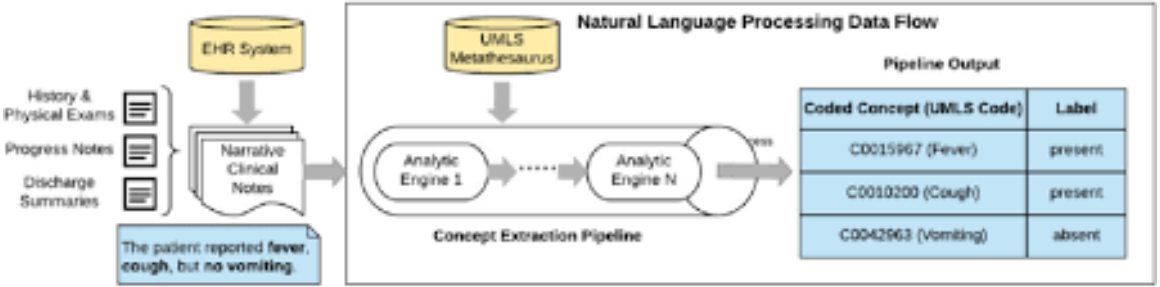
Find all the steps that require classifiers

- Step 1: Section classification. Classify each section of the document to determine which section has the clinical note containing the description of the symptoms.



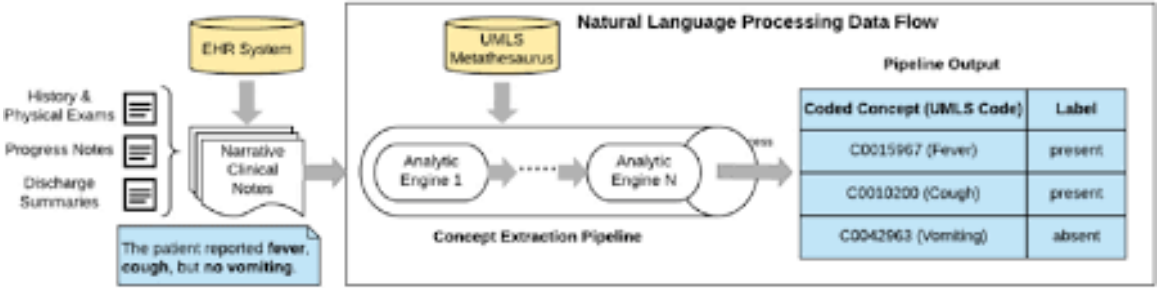
Find all the steps that require classifiers

- Step 2: Concept extraction. For each segment in the sentence, identify whether it's a medical concept. These concepts related to physical symptoms. These are defined relative to a curated medical ontology, here the UMLS ontology.



Find all the steps that require classifiers

- Step 3: Negation classification. We need to find both symptoms that are present and symptoms that are **not** present. For each concept look at the words around it to determine if it's being negated (e.g, "no vomiting"). If yes, the symptom is *absent*; else, the symptom is *present*.



World knowledge: The key challenge for classifiers

- Most classifiers rely on lexical triggers such as the words **great** or **terrible** in a movie review
- But meaning is often conveyed indirectly without such trigger words: "I would rather have my appendix remove than see that movie" This is a negative review but there are no lexical triggers indicating negative sentiment.
- Reporter: "How do you plan to win today?" Manager "What we have to do is go out there and play baseball." Why mention that they're going to play baseball–was there a possibility that they'd play golf instead of baseball? What does the manager really mean here?
- **Pragamatics**: How context contributes to meaning in language, focusing on what speakers intend to communicate beyond the literal meaning of their words

Pragmatic categories

PRAGMATICS

Pragmatics is the study of how context influences how we interpret and make meaning of communication. It is often described as the study of "language in use".

OVERVIEW

Sometimes, the literal meaning of what is said isn't the implied meaning. Pragmatics goes beyond the literal meaning of words and phrases to understand the intended message based on factors like the social situation, relationship between the speakers, cultural context, situational context, and the way the words are said.

EXAMPLES

- Use of metaphor
- Use of irony
- Use of sarcasm
- Use of deixis
- Use of euphemisms
- Jokes and humor
- Use of silence / mis en scene
- Use of hyperbole
- Use of tautologies

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1 Introduction

2 Information Extraction

3 Relation Extraction

4 Coreference Resolution

5 Logical Entailment

6 Intent Classification

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Information Extraction

- Information extraction (IE) turns the unstructured information embedded in texts into structured data,
- You're given the task of determining the relationship (if any) between airline announcements of fare increases and the behavior of their stocks the next day. How can we extract the relevant data from the following article?

> Citing high fuel prices, United Airlines said Friday it has increased fares by $6 per round trip on flights to some cities also served by lower-cost carriers. American Airlines, a unit of AMR Corp., immediately matched the move, spokesman Tim Wagner said. United, a unit of UAL Corp., said the increase took effect Thursday and applies to most routes where it competes against discount carriers, such as Chicago to Dallas and Denver to San Francisco.

Information Extraction

- **Relation extraction**: finding and classifying semantic relation extraction relations among entities mentioned in a text, like child-of (X is the child-of Y), or part-whole or geospatial relations.
- **Event extraction**: the task of finding events in which these event extraction entities participate; for example, in our sample text, the fare increases by United and American and the reporting events *said* and *cite*.
- **Template filling**: find recurring stereotypical events or situations in documents and fill in the template slots. These slot-fillers may consist of text segments extracted directly from the text, or concepts like times, amounts, or ontology entities that have been inferred through additional processing.

Information Extraction

FARE-RAISE ATTEMPT:	LEAD AIRLINE:	UNITED AIRLINES
	AMOUNT:	$6
	EFFECTIVE DATE:	2006-10-26
	FOLLOWER:	AMERICAN AIRLINES

1 Introduction

2 Information Extraction

3 Relation Extraction

4 Coreference Resolution

5 Logical Entailment

6 Intent Classification

7 Argumentation Mining

Relation Extraction

- **Relation extraction**: Finding and classifying semantic relation extraction relations among entities mentioned in a text, like child-of (X is the child-of Y), or part-whole or geospatial relations.
- Lets assume that we have detected the named entities in our sample text and would like to discern the relationships that exist among the detected entities:

> Citing high fuel prices, [$_{ORG}$ **United Airlines**] said [$_{TIME}$ **Friday**] it has increased fares by [$_{MONEY}$ **$6**] per round trip on flights to some cities also served by lower-cost carriers. [$_{ORG}$ **American Airlines**], a unit of [$_{ORG}$ **AMR Corp.**], immediately matched the move, spokesman [$_{PER}$ **Tim Wagner**] said. [$_{ORG}$ **United**], a unit of [$_{ORG}$ **UAL Corp.**], said the increase took effect [$_{TIME}$ **Thursday**] and applies to most routes where it competes against discount carriers, such as [$_{LOC}$ **Chicago**] to [$_{LOC}$ **Dallas**] and [$_{LOC}$ **Denver**] to [$_{LOC}$ **San Francisco**].

Relation Extraction

- The text tells us, for example, that *Tim Wagner* is a spokesman for *American Airlines*, that *United* is a unit of *UAL Corp.*, and that *American* is a unit of *AMR*. These binary relations are instances of more generic relations such as part-of or employs that are fairly frequent in news-style texts.
- **Ontologies** are databases that encode these relations relations

Relations	Types	Examples
Physical-Located	PER-GPE	**He** was in **Tennessee**
Part-Whole-Subsidiary	ORG-ORG	**XYZ**, the parent company of **ABC**
Person-Social-Family	PER-PER	**Yoko**'s husband **John**
Org-AFF-Founder	PER-ORG	**Steve Jobs**, co-founder of **Apple**...

Relation Extraction

- The UMLS (Unified Medical Language System) is one of the more famous ontologies.
- 134 broad subject categories, entity types, and 54 relations between the entities

Entity	Relation	Entity
Injury	disrupts	Physiological Function
Bodily Location	location-of	Biologic Function
Anatomical Structure	part-of	Organism
Pharmacologic Substance	causes	Pathological Function
Pharmacologic Substance	treats	Pathologic Function

- Given a medical sentence like this one:

  Doppler echocardiography can be used to diagnose left anterior descending artery stenosis in patients with type 2 diabetes

  we could thus extract the UMLS relation: **Echocardiography, Doppler** Diagnoses **Acquired stenosis**

1 Introduction

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Coreference Resolution

- An important component of language processing is knowing who is being talked about in a text. Consider the following passage:

> Victoria Chen, CFO of Megabucks Banking, saw her pay jump to $2.3 million, as the 38-year-old became the company's president. It is widely known that she came to Megabucks from rival Lotsabucks.

- Each of the underlined phrases in this passage is used by the writer to refer to a person named *Victoria Chen*. We call linguistic expressions like *her* or *Victoria Chen* **mentions** or referring expressions, and the discourse entity that is referred referent to (Victoria Chen) the **referent**.
- Two or more referring expressions that are used to refer to the same discourse entity are said to **corefer**; thus, *Victoria Chen* and *she* corefer

Coreference Resolution

A dialogue system that has just told the user "There is a 2pm flight on United and a 4pm one on Cathay Pacific" must know which flight the user means by Ill take the **second one**.

- A question answering system that uses Wikipedia to answer a question about Marie Curie must know who **she** was in the sentence "She was born in Warsaw."

1 Introduction

2 Information Extraction

3 Relation Extraction

4 Coreference Resolution

5 Logical Entailment

6 Intent Classification

7 Argumentation Mining

Logical Entailment

- Also known as **Natural Language Inference (NLI)** or **Recognizing Textual Entailment (RTE)**.
- It is the task of determining the semantic relationship between two text segments: a **Premise (P)** and a **Hypothesis (H)**.
- The goal is to classify the relationship into one of three categories:
  1. **Entailment** ($\mathbf{P} \Rightarrow \mathbf{H}$): If the premise is true, the hypothesis must also be true.
  2. **Contradiction** ($\mathbf{P} \Rightarrow \neg\mathbf{H}$): If the premise is true, the hypothesis must be false.
  3. **Neutral**: The premise neither guarantees the truth nor the falsehood of the hypothesis.

Logical Entailment

Definition: Entailment

The truth of the Premise **P guarantees** the truth of the Hypothesis **H**.

Example 1: Lexical Hypernymy

P: A Labrador is fetching a ball.
H: A **dog** is fetching a ball.
Relationship:

Example 2: Syntactic Variation

P: The company was acquired by a rival firm last month.
H: A rival firm acquired the company last month.
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Example 2: Syntactic Variation

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H: A rival firm acquired the company last month.
Relationship: Entailment (Same meaning via **Active/Passive** transformation).

Logical Entailment

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Example 1: Explicit Negation

P: All students passed the final exam.
H: **No** students passed the final exam.
Relationship:

Example 2: World Knowledge/Semantic Conflict

P: A person is swimming in the ocean.
H: The person is **completely dry**.
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Relationship: Contradiction (Swimming implies getting wet; this is **common sense reasoning**).

Logical Entailment

Definition: Neutral

The Premise **P neither confirms nor refutes** the Hypothesis **H**.

Example 1: Additional Information/Unrelated

P: The concert was well attended.
H: The lead singer wore a blue shirt.
Relationship:

Example 2: Possibility vs. Necessity

P: John saw a dog.
H: John saw a Golden Retriever.
Relationship:

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Example 2: Possibility vs. Necessity

P: John saw a dog.
H: John saw a Golden Retriever.
Relationship: Neutral (It could be a Golden Retriever, but P doesn't guarantee it; it could be any dog).

Logical Entailment

- **Fundamental Task**: Serves as a benchmark for general **Natural Language Understanding (NLU)**.
- **Question Answering (QA)**:
  - Used to verify if a candidate answer entails the question (when rephrased as a declarative statement).
- **Summarization**:
  - Used to ensure that sentences in the summary are entailed by the source document (i.e., maintaining factual consistency).
- **Fact Checking/Verification**:
  - Check if a claim (Hypothesis) is entailed or contradicted by a piece of evidence (Premise).

1 Introduction

2 Information Extraction

3 Relation Extraction

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What is the user's intended purpose

- Classic approaches to task-oriented dialogue system (reservation systems,etc.) begin with an intent classification step.

User utterances	Intent Label
Book a flight from LA to Madrid.	Book flight
Can you get me a table at Steve's?	Restaurant reservation
Book Delta ticket Madison to Atlanta.	Book flight
Schedule me a table at Red Lobster.	Restaurant reservation
...	...
Can you tell me the name of this song?	**Open**
Look up the calories in an apple.	**Open**

- Classic approaches to task-oriented dialogue system (reservation systems,etc.) begin with an intent classification step.
- Arguments required to fulfill the user's intent are then extracted.

Rule-Based Approaches (1960s – 1990s)

- **Mechanism:** Relies on hard-coded patterns, regular expressions (Regex), and keyword matching.
- **Key Example:** *ELIZA* (1966) used pattern matching to simulate therapy.
- **Strengths:**
  - High precision for specific, expected keywords.
  - No training data or GPU required.
  - Completely transparent and explainable.
- **Weaknesses:**
  - **Brittle:** Fails if the user uses a synonym or a typo.
  - **Maintenance:** Scale becomes impossible as the number of "rules" grows.
  - No understanding of *context* or *semantics*.

Classical Machine Learning (2000s – 2010s)

- **Mechanism:** Uses statistical models trained on labeled datasets.
- **Algorithms:** Support Vector Machines (SVM), Naive Bayes, Random Forests.
- **Feature Engineering:** Relies on "Bag of Words" (BoW) or TF-IDF.

The Process

Text → Preprocessing (Stemming/Stopwords) → Vectorization → Classifier → Intent Label.

- **Improvement:** Better generalization than rules; handles variations in phrasing better.

BERT-Based Classification (2018 – Present)

- **Mechanism: B**idirectional **E**ncoder **R**epresentations from **T**ransformers.
- **Core Innovation:** Understands word meaning based on surrounding context (e.g., "bank" in "river bank" vs. "bank account").

Feature	BERT Approach
Architecture	Encoder-only Transformer
Input	[CLS] token + sentence
Training	Fine-tuning on labeled intent data
Performance	State-of-the-Art for NLU tasks

- **Pros:** Deep semantic understanding; handles complex syntax.
- **Cons:** Requires a significant labeled dataset for fine-tuning.

LLM-Based Approaches (2023 – Future)

- **Mechanism:** Using generative models (GPT-4, Claude, Llama 3) via prompting.
- **Key Paradigm Shift: In-Context Learning**.

Zero-Shot	"Classify this text into: [Refund, Tech Support, Greeting]."
Few-Shot	Providing 3-5 examples in the prompt to "teach" the model.
Reasoning	LLMs can explain *why* an intent was chosen (Chain-of-Thought).

- **Benefit:** No training data required to start; handles "Out-of-Scope" queries gracefully.
- **Trade-off:** Higher latency and API costs compared to a small BERT model.

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3 Relation Extraction

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What is Argumentation Mining (AM)?

- **Definition**: An NLP subfield focused on the automatic identification and extraction of argumentative structures from natural language text.
- **Goal**: To transform unstructured text (e.g., essays, debates, reviews) into a structured, machine-readable argument graph.
- It goes beyond simple sentiment analysis to understand **why** a person holds a certain opinion.

The Importance of AM

- **NLU Benchmark**: A major step towards true Natural Language Understanding (NLU), as human communication is often argumentative.
- **Decision Making**: Provides structured input for systems that aggregate and compare arguments (e.g., policy support).
- **Critical Thinking**: Used in educational settings to assess and provide feedback on students' argumentation skills.

The Argumentation Mining Pipeline (Subtasks)

Argumentation Mining is typically broken down into four sequential subtasks:

1. **Argument Component Identification (ACI)**
   - Segmenting the text into elementary units that might be argumentative.

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3. **Argumentative Relation Identification (ARI)**
   - Determining which components are linked (e.g., P and C).
4. **Argumentative Relation Classification (ARC)**
   - Classifying the type of relationship (Support or Attack/Rebuttal).

Key Argument Components

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Claim (Conclusion) The main stance or assertion the author is trying to prove.

- *Example*: "Smoking should be banned in all public places."

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Premise (Data/Evidence) A statement offered as a reason or evidence to support the Claim.

- *Example*: "Second-hand smoke causes serious health risks to non-smokers."

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- *Example*: "Second-hand smoke causes serious health risks to non-smokers."

Warrant (Toulmin Model) The (often implicit) general rule or principle that connects the Premise to the Claim.

- *Example*: *(Implicit)* "We should ban activities that cause serious health risks to the public."

A Complete Argument Example

Text Snippet

The voting age should be lowered to 16.

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The voting age should be lowered to 16. Teenagers can legally work and pay taxes, so they are active participants in the economy.

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The voting age should be lowered to 16. Teenagers can legally work and pay taxes, so they are active participants in the economy. Furthermore, many 16-year-olds are already educated in civics, making them informed voters.

A Complete Argument Example

Text Snippet

The voting age should be lowered to 16. Teenagers can legally work and pay taxes, so they are active participants in the economy. Furthermore, many 16-year-olds are already educated in civics, making them informed voters. However, some argue that **lowering the age would increase voter volatility** because **young people lack life experience**.

- **Main Claim (C)**: The voting age should be lowered to 16.
- **Premise 1 (P_1)**: Teenagers can legally work and pay taxes...
- **Premise 2 (P_2)**: Many 16-year-olds are already educated in civics...
- **Counter-Claim (C')**: lowering the age would increase voter volatility.
- **Counter-Premise (P')**: young people lack life experience.

Argument Mining I/O

- Current Study
- Claim
- Citation
- Hypothesis
- Supports
- Opposes

Stop-signs are a valuable part of traffic safety, which are often ignored, resulting in tragic crashes. In terms of total intersection crashes and fatalities between 1997 and 2004, intersection controlled by stop-signs had the most crashes and fatalities. Study This study provides valuable information that can be used toward programs for the increase of the proper obedience to stop-sign laws, which will contribute to the reduction of the number of intersection crashes. Claim Stop-signs indicate that the driver must come to a complete stop before the sign and check for oncoming and opposing traffic before Temporal proceeding on. For a stop to be considered complete the car must completely stop moving. Four-way stop intersections have a stop-sign placed on all four directions. All cars must stop before Temporal passing through the intersection and Expansion then Temporal the car, which stops first is given the right of way to pass through the intersection. Traffic activity is determined by the number of cars during a given period of time, higher traffic activity means that there are more cars.

The purpose of this activity is to determine the effect of traffic activity on the likelihood of the drivers making a stop-sign violation. Cite kerstedt & Kecklund (2001) did a similar study on traffic accident risk and found a relationship between time of day, gender, and age on the risk of highway accidents. In the current study however Comparison, it is local urban traffic which is studied and Expansion it adds in the factor of traffic activity. Also Expansion, there is much prior research on time of day as related to tiredness, but Comparison in this study it is used in relation to traffic activity. While Comparison there are many studies on the internal factors of driving risk, there is less on outside factors which the drivers have no control over, such as traffic. It is important to study traffic because Contingency it greatly affects how one drives, and Expansion this study is attempting to increase the understanding of the relationship between the two.

Stop The first hypothesis was: H Contingency it is a high activity time of day at an intersection then Contingency, there will be a higher ratio of complete stops made than during a low activity time at the intersection. Stop The second hypothesis was: H Contingency there is a busy intersection then Contingency, there will be a higher ratio of complete stops made than at an intersection that is less busy. Stop So Contingency essentially, it was expected that when Temporal there was a higher traffic activity level, either due to location or time of day, there were to be less stop-sign violations. Claim There have been many studies which indicate that people do drive differently at different times of day and Expansion that it does have an impact on driving risk. Cite Reimer et al (2007) found that time of day did influence driving speed, reaction time, and speed variability measures. All of which are factors in driving risk, impacting the likelihood of a traffic violation, such as running a stop sign. Cite Otmani et al (2005) study supports the second hypothesis with their finding that young drivers faced a significant decrease in alertness while in low traffic conditions. This decrease in alertness can then Temporal negatively impact a driver's judgment indicating a greater chance that he\she will have a traffic violation. Stop However Comparison, McGarva & Zeiner (2000) oppose the second hypothesis because Contingency they found that provoked driver aggression through honking horns increased the rate of acceleration at a stop sign. Drivers have a greater chance of encountering a provoking driver during times of higher traffic, which by influencing more aggressive driving can led to more traffic violations. Stop However Comparison, this did not have a great effect on the formation of the second hypothesis because of the

Next class: March 17

Sequence Modeling Tasks

- Named Entity Recognition
- Syntactic parsing
- Semantic parsing

Reading

- Jurafsky & Martin Chapter 17: Sequence Labeling for Parts of Speech and Named Entities
- Jurafsky & Martin Chapter 18:Context-Free Grammars and Constituency Parsing
- Jurafsky & Martin Chapter 21: Semantic Role Labeling